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Solutions

To receive full credit for a problem you must show all necessary work.

1. Determine whether the series converges or diverges. If it converges, find the sum.

$$2 - \frac{2}{3} + \frac{2}{9} - \frac{2}{27} + \frac{2}{81} - \frac{2}{243} + \dots = \sum_{n=1}^{\infty} 2 \left(-\frac{1}{3}\right)^{n-1}$$

It is a geometric series $\sum_{n=1}^{\infty} a \cdot r^{n-1}$ with $a=2$ and $r=-\frac{1}{3}$

$|r| = \frac{1}{3} < 1$. So the series converges,

and the sum is $\frac{a}{1-r} = \frac{2}{1-(-\frac{1}{3})} = \frac{3}{2}$

2. Determine whether the series $\sum_{n=1}^{\infty} \frac{2+7^n}{\pi+3^n}$ converges.

$$\textcircled{1} \quad \frac{a_n}{b_n} = \frac{\frac{2+7^n}{\pi+3^n}}{\frac{7^n}{3^n}} = \frac{2+7^n}{\pi+3^n} \cdot \frac{3^n}{7^n} = \frac{2 \cdot 3^n + 21^n}{\pi \cdot 7^n + 21^n} = \frac{2 \cdot (\frac{1}{7})^n + 1}{\pi (\frac{1}{3})^n + 1} \rightarrow 1 \quad \text{when } n \rightarrow \infty$$

② Since $\sum b_n = \sum \left(\frac{7}{3}\right)^n$ is a geometric series with $|r| = \frac{7}{3} > 1$
So $\sum b_n$ is divergent.

③ By limit comparison test, the series $\sum_{n=1}^{\infty} \frac{2+7^n}{\pi+3^n}$ is divergent.

3. Determine whether the series $\sum_{n=1}^{\infty} (-1)^n \frac{2}{\sqrt{3n-1}}$ converges.

Alternating series.

① $b_n = \frac{2}{\sqrt{3n-1}} > 0$

② b_n is decreasing.

③ $\lim_{n \rightarrow \infty} \frac{2}{\sqrt{3n-1}} = 0$

$f(x) = \frac{2}{\sqrt{3x-1}}$. $f'(x) = 2(-\frac{1}{2})(3x-1)^{-\frac{3}{2}} \cdot 3 < 0$
when $x > 1$.

So the alternating series $\sum_{n=1}^{\infty} (-1)^n \frac{2}{\sqrt{3n-1}}$ is convergent.

4. Determine whether the series $\sum_{n=0}^{\infty} \frac{10^n}{n!}$ converges.

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{10^{n+1}}{(n+1)!} \cdot \frac{n!}{10^n} = \frac{10}{n+1} \rightarrow 0 \text{ when } n \rightarrow \infty.$$

So, by ratio test, $\sum_{n=0}^{\infty} \frac{10^n}{n!}$ is absolutely convergent.

hence, convergent.

5. Find the radius of convergence and interval of convergence of the series $\sum_{n=0}^{\infty} \frac{x^n}{n2^n}$.

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x^{n+1}|}{(n+1)2^{n+1}} \cdot \frac{n \cdot 2^n}{|x^n|} = \frac{n \cdot |x|}{(n+1)2} \rightarrow \frac{|x|}{2} \text{ when } n \rightarrow \infty.$$

So, by ratio test, $\sum_{n=0}^{\infty} \frac{x^n}{n \cdot 2^n}$ is absolutely convergent when $\frac{|x|}{2} < 1$, $|x| < 2$
it is divergent when $|x| > 2$.

- When $x=2$, $\sum_{n=0}^{\infty} \frac{1}{n}$ is divergent.
- When $x=-2$, $\sum_{n=0}^{\infty} (-1)^n \frac{1}{n}$ is divergent.

So, the radius of convergence is $R=2$.

the interval of convergence is $-2 < x < 2$ or $(-2, 2)$

6. Find a power series representation for the function $f(x) = \frac{x}{(2-x)^2}$ and determine the interval of convergence.

$$\cdot \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad |x| < 1.$$

$$\cdot \frac{1}{2-x} = \frac{1}{2} \left(\frac{1}{1-\frac{x}{2}} \right) = \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{x}{2} \right)^n \quad \left| \frac{x}{2} \right| < 1, \quad |x| < 2.$$

$$\cdot \left(\frac{1}{2-x} \right)' = \frac{1}{(2-x)^2} \quad \text{So } \frac{1}{(2-x)^2} = \sum_{n=1}^{\infty} \frac{n \cdot x^{n-1}}{2^{n+1}} \quad |x| < 2$$

$$\cdot \text{So } \frac{x}{(2-x)^2} = \sum_{n=1}^{\infty} \frac{n \cdot x^n}{2^{n+1}}$$

the interval of convergence
is $-2 < x < 2$
or $(-2, 2)$

7. Find the sum $\sum_{n=0}^{\infty} (-1)^n \frac{(\pi/3)^{2n}}{(2n)!}$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \quad R = \infty$$

$$S_0 = \sum_{n=0}^{\infty} (-1)^n \frac{(\pi/3)^{2n}}{(2n)!} = \cos \frac{\pi}{3} = \frac{1}{2}$$

8. (Multi-choice) Determine whether the series is convergent or divergent.

(1.) $\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$ (A) absolutely convergent. (B) conditionally convergent. (C) divergent.

(2.) $\sum_{n=1}^{\infty} \frac{1000}{\sqrt{n^2+1}}$
 (A) absolutely convergent. (B) conditionally convergent. (C) divergent.

(3.) $\sum_{n=1}^{\infty} (-1)^n \frac{1000}{\sqrt{n^2+1}}$
 (A) absolutely convergent. (B) conditionally convergent. (C) divergent.

(4.) $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n^2}\right) (-1)^n$ (A) absolutely convergent. (B) conditionally convergent. (C) divergent.